Ivo Raimondi

Email address: ivo40527@gmail.com | **Website:** ivoraimar.github.io/ | **Website:** github.com/ivoraimar | **LinkedIn:** Ivo Raimondi | **Address:** Seville, Spain (Home)

WORK EXPERIENCE

STUDENT ASSISTANT — SOFTWARE & EDGE COMPUTING – TECHNICAL UNIVERSITY OF MUNICH (TUM) – CHAIR OF CONNECTED MOBILITY – 15/01/2026 – 30/09/2026 – MUNICH, GERMANY

Website: www.oakestra.io | **Link** : https://github.com/oakestra

- Contributed as a maintainer to **Oakestra**, an open-source orchestration framework for distributed edge computing systems.
- Developed and improved **Python/Flask backend APIs** for service and network management.
- Worked on the platform's **observability infrastructure**, including structured logging, metrics collection, Grafana dashboards, and alerting using Prometheus, Loki, and Docker Compose.
- Modernized and extended the **Angular dashboard**, including a major framework upgrade and new features.
- Improved deployment tooling, infrastructure configurations, and technical documentation for distributed edge environments.

EDUCATION AND TRAINING

30/09/2026 – CURRENT Seville, Spain

MASTER'S DEGREE IN INTELLIGENT SYSTEMS IN ENERGY AND TRANSPORT (SMART CITIES) University of Seville

- Specialization in Smart Cities, focusing on intelligent infrastructure, industrial software, IoT, and connected systems.
- Coursework covering Industrial Software Development, Industrial Communications, Industrial Information Systems, Advanced Sensor Networks, and Intelligent Data Processing.
- Applications in Smart Grids, Intelligent Transportation Systems, Smart Buildings, and Energy Efficiency.

Level in EQF EQF level 7

12/10/2025 – 06/08/2026 Munich, Germany

ERASMUS+ EXCHANGE IN INFORMATICS Technical University of Munich (TUM)

- Advanced coursework in Modern C++, Software Engineering, Transaction Systems, Edge Computing & IoT, and Quantum Networks.
- Developed a team-based IoT/Edge rover using Raspberry Pi 5 and ESP32, integrating MQTT communication, telemetry, media streaming, and cloud services.

Thesis Evaluation of Live Migration Strategies at the Edge

13/09/2022 – 06/08/2026 Seville, Spain

BACHELOR'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING (SOFTWARE ENGINEERING SPECIALIZATION) University of Seville

- Software engineering, requirements analysis, software architecture, design patterns, testing and Agile development.
- Computer science foundations, including algorithms, data structures, operating systems, computer networks and distributed systems.
- Developed team-based software projects using Java/Spring Boot, Python, JavaScript/React and relational databases.

Level in EQF EQF level 6

PROJECTS

01/03/2026 – 16/07/2026

Evaluation of Live Migration Strategies at the Edge

- Designed and implemented a benchmarking framework to evaluate service migration strategies in resource-constrained edge environments.
- Evaluated CRIU-based cold, pre-copy, and post-copy migration, alongside WebAssembly migration experiments.
- Automated experiments under varying network conditions, measuring downtime, migration time, data transfer, and resource overhead.

Technologies: Linux, Python, CRIU, Podman, WebAssembly, Terraform, Multipass.

Link https://github.com/Ivoraimar/Evaluation-of-Live-Migration-Strategies-at-the-Edge

13/10/2025 – 20/02/2026

Edge Computing & IoT Rover

- Collaborated on the development of a distributed IoT rover integrating Raspberry Pi 5 and ESP32 devices.
- Implemented and integrated components for MQTT communication, telemetry, video streaming, and cloud/edge services.
- Worked with embedded hardware, networking, and software components in a team-based engineering project.

Technologies: Raspberry Pi 5, ESP32, Python, MQTT, Linux, IoT, Edge Computing.

Link https://github.com/orgs/TUM-Edge-IoT/repositories

SKILLS

Teamwork | Agile development | Problem-solving

Digital Skills

Python (computer programming) | TypeScript | JavaScript | SQL | Java (computer programming) | Angular | Grafana | Linux | Prometheus | Git | Edge Computing | GitHub | Docker | Distributed Systems

LANGUAGE SKILLS

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

CERTIFICATIONS

University of Cambridge, 29/07/2023

C1 Advanced (CAE) – Cambridge English

Cambridge C1 Advanced (CAE) certification, demonstrating advanced English proficiency (CEFR Level C1) in reading, writing, listening, and speaking